ITDS 124 Mathematics of Computing

Chapter 2.4
Mathematical Induction

การอุปนัยเชิงคณิตศาสตร์

Created by : Rawesak Tanawongsuwan
rawesak.tan@mahidol.ac.th

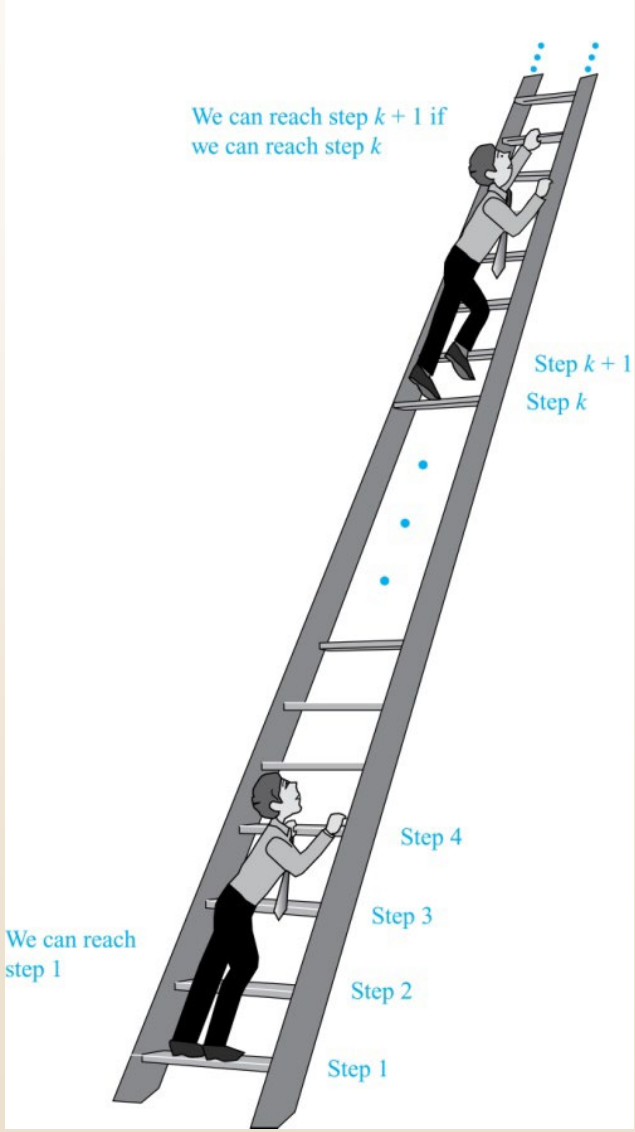
Note on this lecture

- This lecture has been developed from several textbooks including
  - Discrete Mathematical Structures by Bernard Kolman, Robert Busby, et al.
  - Discrete Mathematics and Its Applications by Kenneth Rosen
  - Discrete Mathematics: An Open Introduction by Oscar Levin

Goal for this topic

- Many mathematical statements state that a property is true for all positive integers. Examples of such statements are that
  - For every positive integer n: $n! \leq n^n$.
  - For every positive integer n: $n^3 - n$ is divisible by 3.
  - A set with n elements has 2^n subsets;
  - The sum of the first n positive integers is $\frac{n(n+1)}{2}$.
- A goal of this lecture is to introduce mathematical induction, which is used to prove results of this kind.

Climbing an Infinite Ladder



- Suppose we have an infinite ladder:

  (1) We can reach the first rung of the ladder.

  (2) If we can reach a particular rung of the ladder, then we can reach the next rung.

- From (1), we can reach the first rung.
- Then by applying (2), we can reach the second rung. Applying (2) again, the third rung. And so on.
- We can apply (2) any number of times to reach any particular rung, no matter how high up.

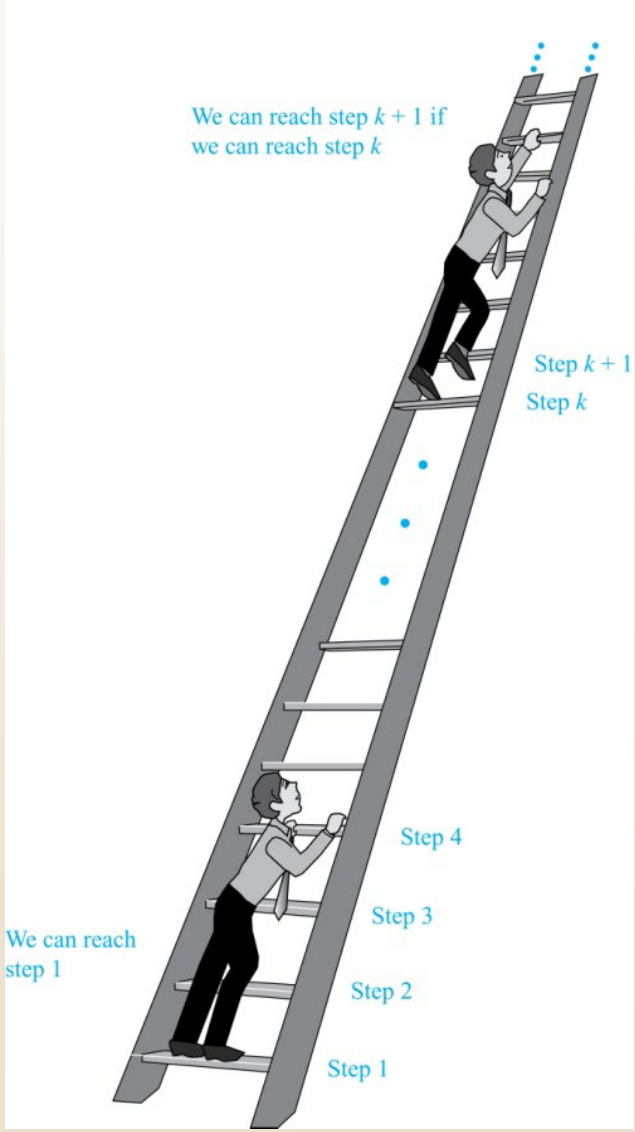
- This example motivates proof by mathematical induction.

Principle of Mathematical Induction

- To prove that $P(n)$ is true for all positive integers n, we complete these steps:
  - **Basis Step**: Show that $P(1)$ is true.
  - **Inductive Step**: Show that $P(k) \rightarrow P(k+1)$ is true for all positive integers k.

    To complete the inductive step, assuming the inductive hypothesis that $P(k)$ holds for an arbitrary integer k, show that $P(k+1)$ must be true.

Principle of Mathematical Induction



- Climbing an Infinite Ladder Example:

  (1) We can reach the first rung of the ladder.

  (2) If we can reach a particular rung of the ladder, then we can reach the next rung.

- To conclude that we can reach every rung by mathematical induction:
  - **Basic Step**: By (1), we can reach rung 1.
  - **Inductive Step**: Assume the inductive hypothesis that we can reach rung k. Then by (2), we can reach rung $k+1$.

    Hence, $P(k) \rightarrow P(k+1)$ is true for all positive integers k. We can reach every rung on the ladder.

Important Points About Using Mathematical Induction

- Mathematical induction can be expressed as the rule of inference

$$\left(P(1) \wedge \forall k\big(P(k) \rightarrow P(k+1)\big)\right) \rightarrow \forall n\, P(n)$$

  where the domain is the set of positive integers.

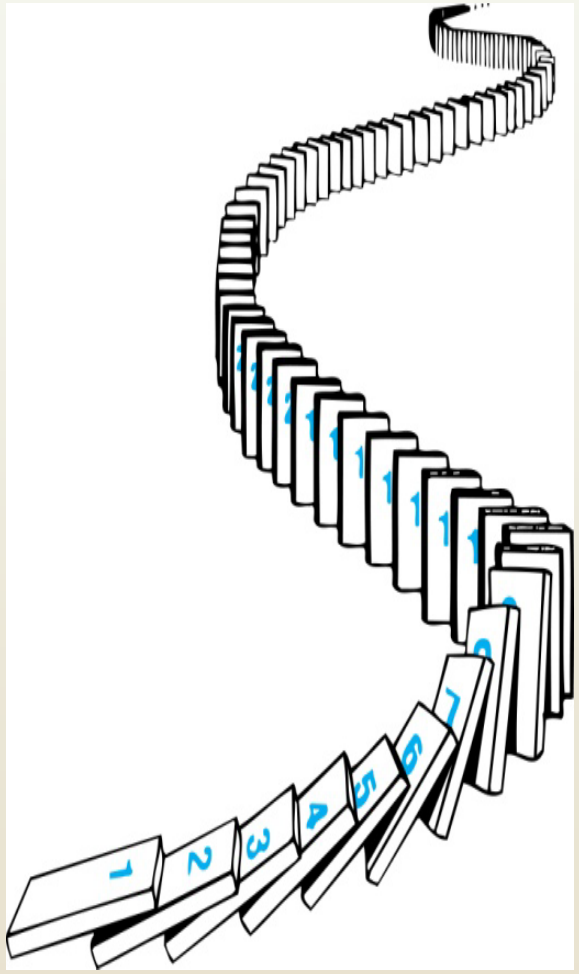
- In a proof by mathematical induction, we don't assume that $P(k)$ is true for all positive integers!

- We show that if we assume that $P(k)$ is true, then $P(k+1)$ must also be true.

- Proofs by mathematical induction do not always start at the integer 1. In such a case, the basis step begins at a starting point *b* where *b* is an integer.

Remembering How Mathematical Induction Works

- Beside the infinite ladder example, we can look at another example of dominos.



- Consider an infinite row of dominoes, labeled $1,2,3,\ldots,n,\ldots$, where each domino is standing up.
- Let $P(n)$ be the proposition that domino n is knocked over.
- If the first domino is knocked over—i.e., if $P(1)$ is true—and if, whenever the kth domino is knocked over, it also knocks the $(k+1)^{st}$ domino over—i.e., if $P(k) \rightarrow P(k+1)$ is true for all positive integers k—then all the dominoes are knocked over.

The Good and the Bad of Mathematical Induction

- The good thing about mathematical induction is that it can be used to prove a conjecture once it has been made (and is true).

- The bad thing about it is that it cannot be used to find new theorems.

- Mathematicians sometimes find proofs by mathematical induction unsatisfying because they do not provide insights as to why theorems are true.
- Many theorems can be proved in many ways, including by mathematical induction.
- Proofs of these theorems by methods other than mathematical induction are often preferred because of the insights they bring.

Examples of Proofs by Mathematical Induction

- Many theorems assert that $P(n)$ is true for all positive integers n, where $P(n)$ is a propositional function.

- Mathematical induction is a technique for proving theorems of this kind.
- In other words, mathematical induction can be used to prove statements of the form $\forall n\, P(n)$, where the domain is the set of positive integers.

- Mathematical induction can be used to prove a wide variety of theorems, for example, summation formulae, inequalities, identities for combinations of sets, divisibility results, theorems about algorithms, the correctness of computer programs and algorithms, etc.

Example 1: Proving summation formula

- Show that if n is a positive integer, then $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$.

Solution: Let $P(n)$ be the proposition that $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$.

- **BASIS STEP**: $P(1)$ is true, because _____
- **INDUCTIVE STEP**: Assume that $P(k)$ is true. That is, we assume that _____.

  Under this assumption, we must show that $P(k+1)$ is true, or _____.

  When we add $k + 1$ to both sides of the equation in $P(k)$, we obtain

  $1 + 2 + \cdots + k + (k + 1) =$ _____ $=$ _____ $=$ _____

  We show that $P(k+1)$ is true under the assumption that $P(k)$ is true. This completes the inductive step.
- We have completed the basis step and the inductive step, so by mathematical induction we know that $P(n)$ is true for all positive integers n. That is, we have proven that $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$ for all positive integers n.

Example 2: Conjecturing and Proving Correct a Summation Formula

- Conjecture a formula for the sum of the first n positive odd integers. Then prove your conjecture using mathematical induction.

Solution:

We will deal with the conjecturing part first.

The sums of the first n positive odd integers for $n = 1,2,3,4,5$ are

$$1 = 1, \qquad 1 + 3 = 4, \qquad 1 + 3 + 5 = 9, \qquad 1 + 3 + 5 + 7 = 16, \qquad 1 + 3 + 5 + 7 + 9 = 25$$

From these values it is reasonable to conjecture that the sum of the first n positive odd integers is n^2, that is, $1 + 3 + 5 + \cdots + (2n - 1) = n^2$.

We need a method to prove that this conjecture is correct, if in fact it is.

So we are going to prove it with mathematical induction.

Continue to the next slide

Example 2: Conjecturing and Proving Correct a Summation Formula

- Conjecture a formula for the sum of the first n positive odd integers. Then prove your conjecture using mathematical induction.

Solution: Let $P(n)$ be the proposition that $1 + 3 + 5 + \cdots + (2n - 1) = n^2$.

BASIS STEP: _____

INDUCTIVE STEP: $P(k) \rightarrow P(k + 1)$ for every positive integer k.

Assume the inductive hypothesis is true and then show that $P(k + 1)$ is also true.

Assuming $P(k)$ is true, it follows that

$$\begin{aligned} 1 + 3 + 5 + \cdots + (2k - 1) + (2k + 1) &= \underline{\qquad\qquad\qquad\qquad} \\ &= \underline{\qquad\qquad\qquad\qquad} \\ &= \underline{\qquad\qquad\qquad\qquad} \end{aligned}$$

This shows that $P(k + 1)$ follows from $P(k)$.

We have now completed both the basis step and the inductive step. Therefore, the sum of the first n positive odd integer is n^2

Example 3: Proving with basis step $P(b), b \neq 1$

- Use mathematical induction to show that $1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$ for all nonnegative integers n.

Solution: Let $P(n)$ be the proposition that $1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$.

BASIS STEP: _____ is true because _____

INDUCTIVE STEP: For the inductive hypothesis, we assume that $P(k)$ is true for an arbitrary nonnegative integer k. That is, we assume that_____. Then we proceed to show that

$$\begin{aligned} 1 + 2 + 2^2 + \cdots + 2^k + 2^{k+1} &= _____ \\ &= _____ \\ &= _____ \\ &= _____ \end{aligned}$$

We have completed the inductive step.

Because we have completed the basis step and the inductive step, by mathematical induction we know that $P(n)$ is true for all nonnegative integers n. That is, $1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$ for all nonnegative integers n.

Example 4: Proving Inequalities

- Use mathematical induction to prove that $n < 2^n$ for all positive integers n.

Solution:

Let $P(n)$ be the proposition that $n < 2^n$.

BASIS STEP: $P(1)$ is true since $1 < 2^1 = 2$.

INDUCTIVE STEP: Assume $P(k)$ holds, i.e.,_____, for an arbitrary positive integer k.

We must show that $P(k+1)$ holds. Since by the inductive hypothesis, _____, it follows that:

_____

We have completed the inductive step.

Because we have completed the basis step and the inductive step. Therefore $n < 2^n$ for all positive integers n.

Example 5: Proving another inequalities

- Use mathematical induction to prove that $2^n < n!$, for every integer $n \geq 4$.

Solution: Let $P(n)$ be the proposition that _____.

BASIS STEP:

INDUCTIVE STEP:

- Note that here the basis step is $P(4)$, since $P(0), P(1), P(2)$, and $P(3)$ are all false.

Example 6: Proving Divisibility Results

- Use mathematical induction to prove that $n^3 - n$ is divisible by 3, for every positive integer n.

Solution:

Example 7: Proving Number of Subsets of a Finite Set

- Use mathematical induction to show that if S is a finite set with n elements, where n is a nonnegative integer, then S has 2^n subsets.

Solution: Let $P(n)$ be the proposition that a set with n elements has 2^n subsets.

Basis Step: $P(0)$ is true, because the empty set has only itself as a subset and $2^0 = 1$.

Inductive Step: Assume $P(k)$ is true, so every set with k elements has 2^k subsets.

To show $P(k+1)$ follows:

Let T be a set with $k+1$ elements. Then $T = S \cup \{a\}$, where $a \in T$ and $S = T - \{a\}$. Hence $|S| = k$.

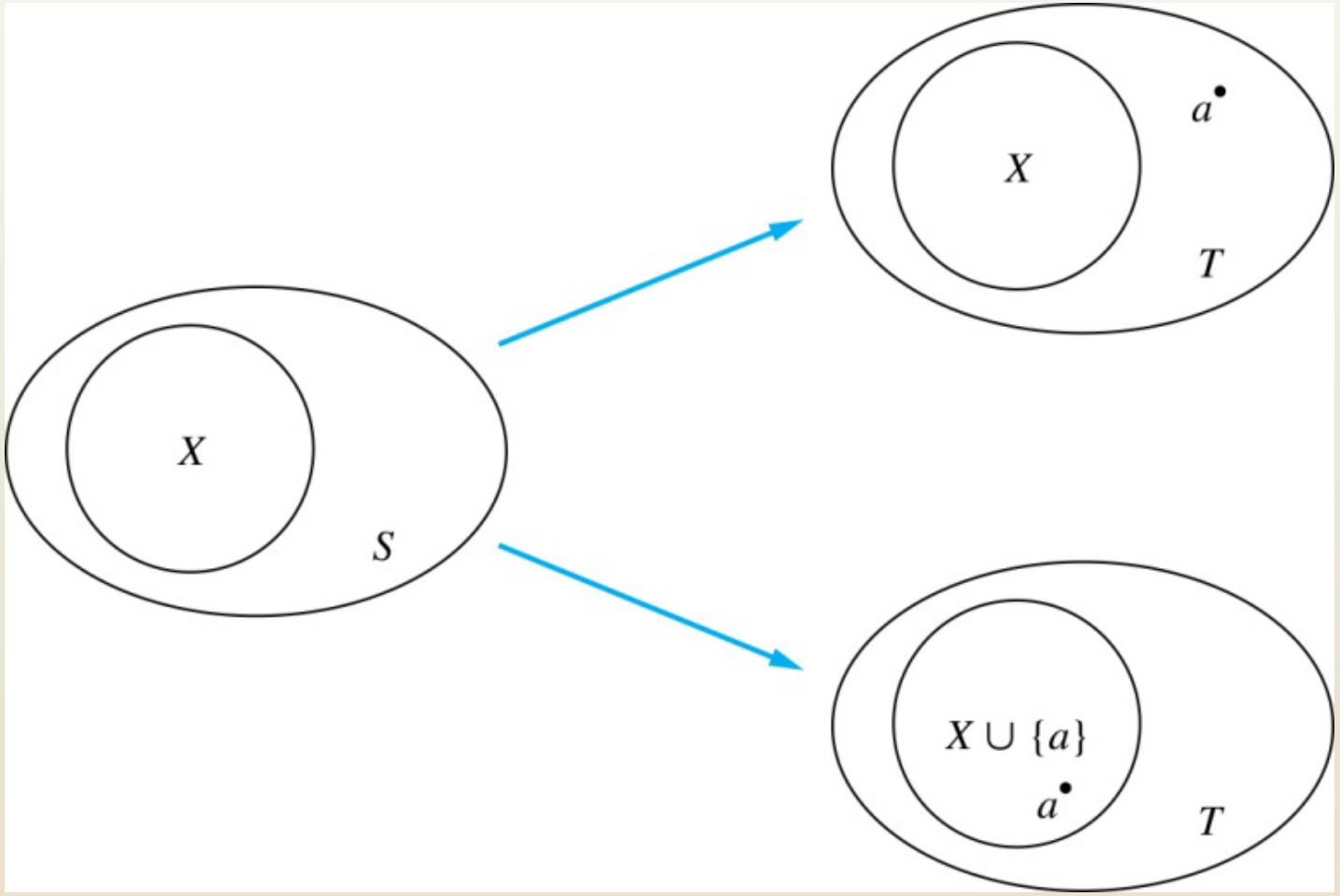
For each subset X of S, there are exactly two subsets of T, i.e., X and $X \cup \{a\}$.

By the inductive hypothesis S has 2^k subsets. Since there are two subsets of T for each subset of S, the number of subsets of T is $2 \cdot 2^k = 2^{k+1}$.

Therefore, we have proved that a set with n elements has 2^n subsets.

Example 7: Proving Number of Subsets of a Finite Set

- Here is the figure to show the concept in the inductive step.



Example 8: Proving De Morgan's Laws

- Use mathematical induction to prove the following generalization of one of De Morgan's laws:

$$\overline{\bigcap_{j=1}^{n} A_j} = \bigcup_{j=1}^{n} \overline{A_j}$$

whenever $A_1, A_2, \ldots, A_n$ are subsets of a universal set U and $n \geq 2$.

Solution: Let $P(n)$ be the identity for n sets.

BASIS STEP: The statement _____ is that _____. This is true because _____.

Example 8: Proving De Morgan's Laws

INDUCTIVE STEP: Assume $P(k)$ is true, so _____ whenever $A_1, A_2, \ldots, A_k$ are subsets of a universal set U and $k \geq 2$.

To show $P(k+1)$, it follows that

This completes the inductive step.

Therefore, we have proved that $\overline{\bigcap_{j=1}^{n} A_j} = \bigcup_{j=1}^{n} \overline{A_j}$ whenever $A_1, A_2, \ldots, A_n$ are subsets of a universal set U and $n \geq 2$.